

Suggested Solutions to 2018 SH2 H2 Mathematics Preliminary Examination Paper 1

1

$$\begin{aligned} u_n - u_{n-1} &= n!(n-2) - (n-1)!(n-3) \\ &= (n-1)!n(n-2) - (n-1)!(n-3) \\ &= (n-1)!(n^2 - 2n - n + 3) \\ &= (n-1)!(n^2 - 3n + 3) \end{aligned}$$

$$\sum_{n=3}^{N+1} [(n-1)!(2n^2 - 6n + 6)].$$

$$2 \sum_{n=3}^{N+1} ((n-1)!(n^2 - 3n + 3))$$

$$= 2 \sum_{n=3}^{N+1} (u_n - u_{n-1})$$

$$= 2 \left[\begin{array}{l} u_3 - u_2 \\ + u_4 - u_3 \\ + u_5 - u_4 \\ + u_6 - u_5 \\ + u_7 - u_6 \\ + u_8 - u_7 \\ + u_9 - u_8 \\ + u_{10} - u_9 \\ + u_{11} - u_{10} \\ + u_{12} - u_{11} \\ + u_{13} - u_{12} \\ + u_{14} - u_{13} \\ + u_{15} - u_{14} \\ + u_{16} - u_{15} \\ + u_{17} - u_{16} \\ + u_{18} - u_{17} \\ + u_{19} - u_{18} \\ + u_{20} - u_{19} \\ + u_{21} - u_{20} \\ + u_{22} - u_{21} \\ + u_{23} - u_{22} \\ + u_{24} - u_{23} \\ + u_{25} - u_{24} \\ + u_{26} - u_{25} \\ + u_{27} - u_{26} \\ + u_{28} - u_{27} \\ + u_{29} - u_{28} \\ + u_{30} - u_{29} \\ + u_{31} - u_{30} \\ + u_{32} - u_{31} \\ + u_{33} - u_{32} \\ + u_{34} - u_{33} \\ + u_{35} - u_{34} \\ + u_{36} - u_{35} \\ + u_{37} - u_{36} \\ + u_{38} - u_{37} \\ + u_{39} - u_{38} \\ + u_{40} - u_{39} \\ + u_{41} - u_{40} \\ + u_{42} - u_{41} \\ + u_{43} - u_{42} \\ + u_{44} - u_{43} \\ + u_{45} - u_{44} \\ + u_{46} - u_{45} \\ + u_{47} - u_{46} \\ + u_{48} - u_{47} \\ + u_{49} - u_{48} \\ + u_{50} - u_{49} \\ + u_{51} - u_{50} \\ + u_{52} - u_{51} \\ + u_{53} - u_{52} \\ + u_{54} - u_{53} \\ + u_{55} - u_{54} \\ + u_{56} - u_{55} \\ + u_{57} - u_{56} \\ + u_{58} - u_{57} \\ + u_{59} - u_{58} \\ + u_{60} - u_{59} \\ + u_{61} - u_{60} \\ + u_{62} - u_{61} \\ + u_{63} - u_{62} \\ + u_{64} - u_{63} \\ + u_{65} - u_{64} \\ + u_{66} - u_{65} \\ + u_{67} - u_{66} \\ + u_{68} - u_{67} \\ + u_{69} - u_{68} \\ + u_{70} - u_{69} \\ + u_{71} - u_{70} \\ + u_{72} - u_{71} \\ + u_{73} - u_{72} \\ + u_{74} - u_{73} \\ + u_{75} - u_{74} \\ + u_{76} - u_{75} \\ + u_{77} - u_{76} \\ + u_{78} - u_{77} \\ + u_{79} - u_{78} \\ + u_{80} - u_{79} \\ + u_{81} - u_{80} \\ + u_{82} - u_{81} \\ + u_{83} - u_{82} \\ + u_{84} - u_{83} \\ + u_{85} - u_{84} \\ + u_{86} - u_{85} \\ + u_{87} - u_{86} \\ + u_{88} - u_{87} \\ + u_{89} - u_{88} \\ + u_{90} - u_{89} \\ + u_{91} - u_{90} \\ + u_{92} - u_{91} \\ + u_{93} - u_{92} \\ + u_{94} - u_{93} \\ + u_{95} - u_{94} \\ + u_{96} - u_{95} \\ + u_{97} - u_{96} \\ + u_{98} - u_{97} \\ + u_{99} - u_{98} \\ + u_{100} - u_{99} \end{array} \right]$$

$$= 2(u_{N+1} - u_2)$$

$$= 2[(N+1)!(N-1) - 2!(0)]$$

$$= 2(N+1)!(N-1)$$

2

Let $OC = \mathbf{c}$.

$$\mathbf{b} = \frac{k\mathbf{a} + \mathbf{c}}{1+k}$$

$$(1+k)\mathbf{b} = k\mathbf{a} + \mathbf{c}$$

$$\mathbf{a} \cdot \mathbf{b}(1+k) = \mathbf{a} \cdot (k\mathbf{a} + \mathbf{c})$$

$$0 = k|\mathbf{a}|^2 + \mathbf{a} \cdot \mathbf{c}$$

$$k|\mathbf{a}|^2 = -|\mathbf{a}||\mathbf{c}|\cos\frac{3\pi}{4}$$

$$k|\mathbf{c}|^2 = -|\mathbf{a}| \cdot \frac{1}{2}|\mathbf{a}|\cos\frac{3\pi}{4} \text{ since } |\mathbf{a}| = 2|\mathbf{c}|$$

$$k = -\frac{1}{2}\cos\frac{3\pi}{4}$$

$$k = \frac{\sqrt{2}}{4}$$

3

$$u = \sin \theta \Rightarrow \frac{du}{d\theta} = \cos \theta$$

$$\int \frac{2 \cos \theta - 3 \sin 2\theta}{1 + \sin^2 \theta} d\theta$$

$$= \int \frac{2 \cos \theta - 6 \sin \theta \cos \theta}{1 + \sin^2 \theta} d\theta$$

$$= \int \left(\frac{2 - 6 \sin \theta}{1 + \sin^2 \theta} \right) \cos \theta d\theta$$

$$= \int \frac{2 - 6u}{1 + u^2} du$$

$$= \int \frac{2}{1 + u^2} du - 3 \int \frac{2u}{1 + u^2} du$$

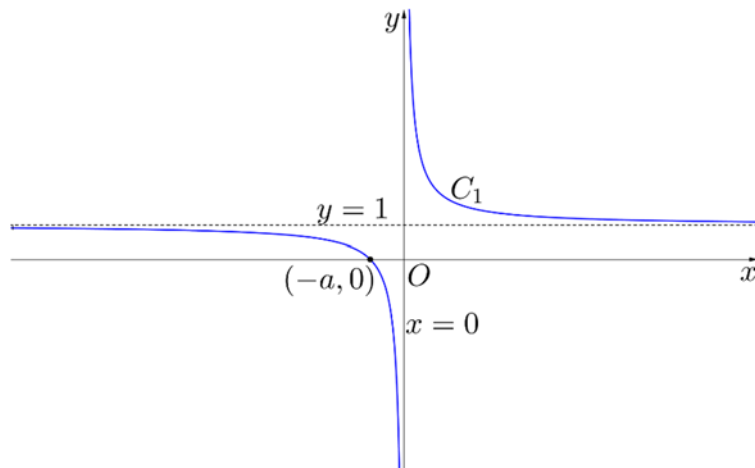
$$= 2 \tan^{-1} u - 3 \ln(1 + u^2) + c$$

$$= 2 \tan^{-1}(\sin \theta) - 3 \ln(1 + \sin^2 \theta) + c$$

$$= 2 \tan^{-1}(\sin \theta) - 3 \ln(1 + \sin^2 \theta) + c$$

4 (i)	Differentiating $9x^2 - y^2 + 3xy - 5 = 0$ w.r.t. x, $18x - 2y \frac{dy}{dx} + 3x \frac{dy}{dx} + 3y = 0$ $2y \frac{dy}{dx} - 3x \frac{dy}{dx} = 18x + 3y$ $\frac{dy}{dx} = \frac{18x + 3y}{2y - 3x}$ Suppose (x, y) is a turning point of C. Then $\frac{dy}{dx} = \frac{18x + 3y}{2y - 3x} = 0 \Rightarrow 18x + 3y = 0 \Rightarrow y = -6x$ Substituting $y = -6x$ into $9x^2 - y^2 + 3xy - 5 = 0$, we get $9x^2 - (-6x)^2 + 3x(-6x) - 5 = 0$ $9x^2 - 36x^2 - 18x^2 - 5 = 0$ $x^2 = \frac{5}{-45} = -\frac{1}{9}$ which is impossible since $x^2 \geq 0$. Therefore C has no turning points.
4 (ii)	At $P(1, -1)$, $\frac{dy}{dx} = \frac{18 - 3}{-2 - 3} = -3$ Then equation of tangent is $\frac{y + 1}{x - 1} = -3 \Rightarrow y = -3x + 2$ Then equation of normal is $\frac{y + 1}{x - 1} = \frac{1}{3} \Rightarrow 3y + 3 = x - 1 \Rightarrow y = \frac{1}{3}x - \frac{4}{3}$
4 (iii)	Area of triangle PQR $= \frac{1}{2} \left(2 + \frac{4}{3} \right) (1)$ $= \frac{5}{3}$

5 (1st
part)



5 (i)

$$x = \frac{a}{t}, y = 1+t \Rightarrow x = \frac{a}{y-1}$$

Substitute $x = \frac{a}{y-1}$ into $y = \sqrt{1 + \frac{x^2}{a^2}}$:

$$y = \sqrt{1 + \left(\frac{\frac{a}{y-1}}{a}\right)^2}$$

$$y^2 = 1 + \frac{1}{(y-1)^2}$$

$$y^2 (y-1)^2 = (y-1)^2 + 1$$

$$y^2 (y-1)^2 - (y-1)^2 - 1 = 0$$

$$(y-1)^2 (y^2 - 1) - 1 = 0$$

Alternatively,

$$y = \sqrt{1 + \frac{\left(\frac{a}{t}\right)^2}{a^2}} = \sqrt{1 + \frac{1}{t^2}}$$

Squaring both sides,

$$y^2 = 1 + \frac{1}{t^2}$$

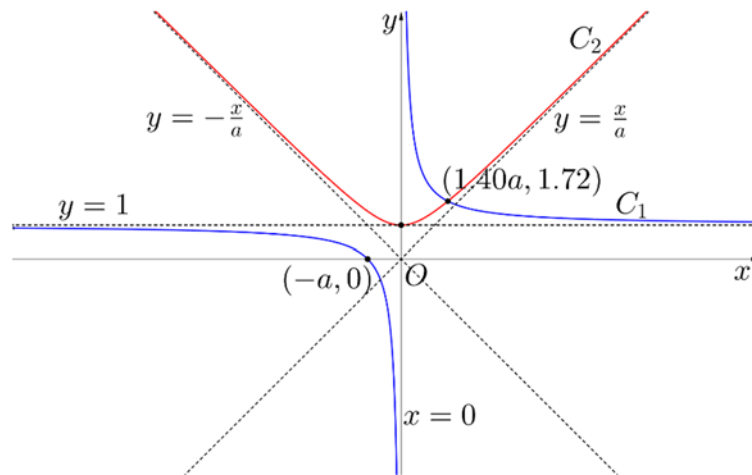
$$y^2 = 1 + \frac{1}{(y-1)^2}$$

$$y^2 - 1 = \frac{1}{(y-1)^2}$$

$$(y^2 - 1)(y-1)^2 = 1$$

$$(y^2 - 1)(y-1)^2 - 1 = 0 \text{ (shown)}$$

5 (ii)



Using GC to solve, $(y-1)^2(y^2-1)-1=0$,

$y = -1.106919$ (rejected) or 1.716673

When $y = 1.716673$, $x = 1.39534a$

$\therefore$ The point of intersection is $(1.40a, 1.72)$.

6 (a)

$$\begin{aligned}2x^2 - \lambda x - 3 &= 2\left(x^2 - \frac{\lambda}{2}x\right) - 3 \\&= 2\left(x^2 + 2\left(-\frac{\lambda}{4}\right)x + \left(-\frac{\lambda}{4}\right)^2\right) - 3 - 2\left(-\frac{\lambda}{4}\right)^2 \\&= 2\left(x - \frac{\lambda}{4}\right)^2 - \frac{\lambda^2 + 24}{8}\end{aligned}$$

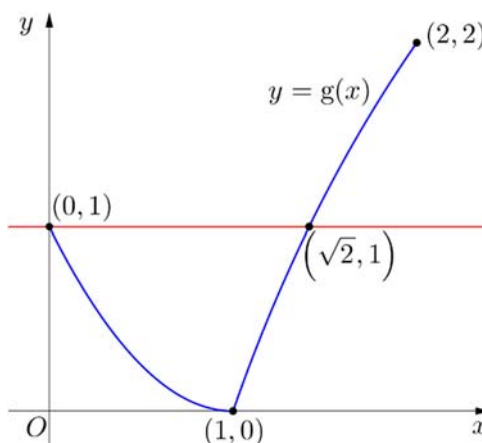
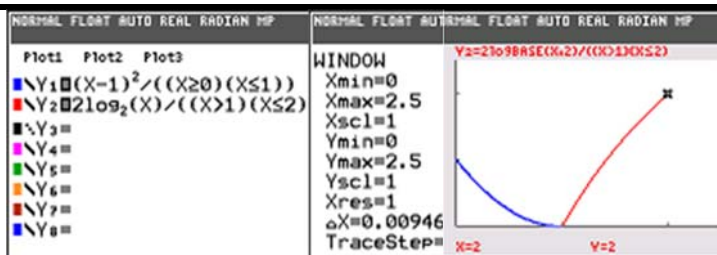
OR

$$\begin{aligned}\frac{d}{dx}(2x^2 - \lambda x - 3) &= 0 \\4x - \lambda &= 0 \\x &= \frac{\lambda}{4}\end{aligned}$$

For f^{-1} to exist, the turning point of $y = 2x^2 - \lambda x - 3$ **cannot** lie in the interval $\frac{7}{4} < x < 5$. Therefore,

$$\frac{\lambda}{4} \leq \frac{7}{4} \text{ or } \frac{\lambda}{4} \geq 5 \Rightarrow \lambda \leq 7 \text{ or } \lambda \geq 20$$

6 (b)(i)



6 (b)(ii)

From the sketch in (b)(i),

Point of intersection between $y = g(x)$ and $y = 1$ occurs at the points $(0, 1)$ and where

$$\begin{aligned} 2 \log_2 x = 1 &\Rightarrow \log_2 x = \frac{1}{2} \\ &\Rightarrow x = 2^{\frac{1}{2}} = \sqrt{2} \end{aligned}$$

Therefore, $1 < g(x) \leq 2 \Rightarrow \sqrt{2} < x \leq 2$

6 (b) (iii)

For $0 \leq x \leq 1$, $g^2(x) = g((x-1)^2) = ((x-1)^2 - 1)^2 = (x^2 - 2x)^2$

Considering the effect of part (b)(ii),

For $1 < x \leq \sqrt{2}$, $g^2(x) = g(2 \log_2 x) = (2 \log_2 x - 1)^2$

For $\sqrt{2} < x \leq 2$, $g^2(x) = g(2 \log_2 x) = 2 \log_2 (2 \log_2 x)$

$$\text{Therefore, } g^2(x) = \begin{cases} (x^2 - 2x)^2 & \text{for } 0 \leq x \leq 1, \\ (2 \log_2 x - 1)^2 & \text{for } 1 < x \leq \sqrt{2}, \\ 2 \log_2 (2 \log_2 x) & \text{for } \sqrt{2} < x \leq 2 \end{cases}$$

7 (a)

$$y + 2 = (x + 1)^{\ln(x+1)}$$

$$\ln(y + 2) = \ln(x + 1)^{\ln(x+1)}$$

$$\ln(y + 2) = (\ln(x + 1))^2$$

$$\frac{1}{y + 2} \frac{dy}{dx} = 2(\ln(x + 1)) \left(\frac{1}{x + 1} \right)$$

$$(x + 1) \frac{dy}{dx} = 2(y + 2) \ln(x + 1) \text{ (shown)}$$

$$(x + 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = \frac{2(y + 2)}{x + 1} + 2 \frac{dy}{dx} \ln(x + 1)$$

$$(x + 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} [1 - 2 \ln(x + 1)] = \frac{2(y + 2)}{x + 1}$$

When $x = 0$,

$$y + 2 = 1$$

$$\therefore y = -1$$

$$\frac{1}{-1 + 2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = 0$$

$$\frac{d^2y}{dx^2} = \frac{2(-1 + 2)}{0 + 1}$$

$$\therefore \frac{d^2y}{dx^2} = 2$$

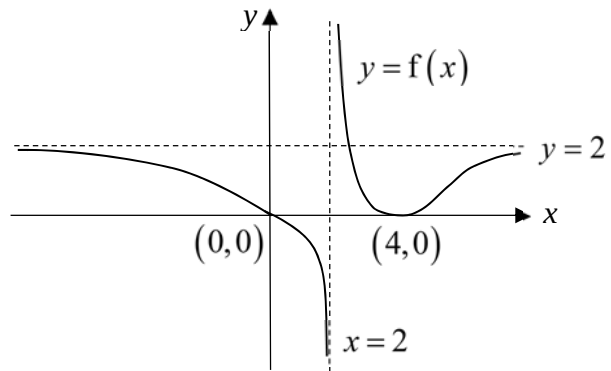
$$y \approx -1 + \frac{x^2}{2!} (2)$$

$$\therefore y = -1 + x^2$$

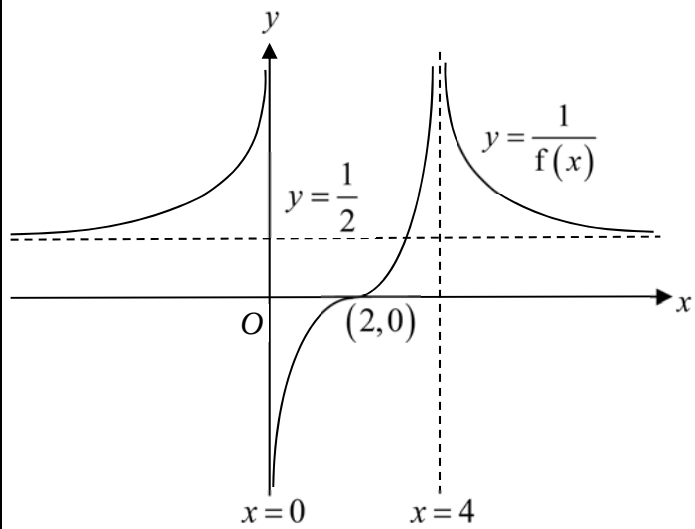
7 (b)(i)	$ \begin{aligned} PQ^2 &= PR^2 + QR^2 - 2PR \cdot QR \cos \angle PRQ \\ &= 5^2 + 6^2 - 2(5)(6) \cos(\angle PRS + \theta) \\ &= 25 + 36 - 60 \left[\cos(\angle PRS) \cos \theta - \sin(\angle PRS) \sin \theta \right] \\ &= 61 - 60 \left(\frac{3}{5} \cos \theta - \frac{4}{5} \sin \theta \right) \\ &= 61 - 36 \cos \theta + 48 \sin \theta \\ \therefore PQ &= (61 - 36 \cos \theta + 48 \sin \theta)^{\frac{1}{2}} \text{ (shown)} \end{aligned} $
7 (b) (ii)	$ \begin{aligned} PQ &= (61 - 36 \cos \theta + 48 \sin \theta)^{\frac{1}{2}} \\ &\approx \left(61 - 36 \left(1 - \frac{\theta^2}{2} \right) + 48\theta \right)^{\frac{1}{2}} \\ &= (25 + 48\theta + 18\theta^2)^{\frac{1}{2}} \\ &= 5 \left(1 + \frac{48}{25}\theta + \frac{18}{25}\theta^2 \right)^{\frac{1}{2}} \\ &\approx 5 \left(1 + \frac{1}{2} \left(\frac{48}{25}\theta + \frac{18}{25}\theta^2 \right) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!} \left(\frac{48}{25}\theta \right)^2 \right) \\ &= 5 \left(1 + \frac{24}{25}\theta + \frac{9}{25}\theta^2 - \frac{288}{625}\theta^2 \right) \\ &= 5 \left(1 + \frac{24}{25}\theta - \frac{63}{625}\theta^2 \right) \\ &= 5 + \frac{24}{5}\theta - \frac{63}{125}\theta^2, \quad p = \frac{24}{5}, \quad q = -\frac{63}{125} \end{aligned} $

8 (a) (i)	1. Reflect graph about the x-axis 2. Scale graph parallel to the y-axis by factor of $(6 + a)$ 3. Translate graph along/in the negative y-direction by 2 units
8 (a) (ii)	$y = \frac{2x + a}{3 - x}$ $= -\frac{2x + a}{x - 3}$ $= -2 - \frac{6 + a}{x - 3}$ $\frac{dy}{dx} = -\frac{6 + a}{(x - 3)^2}(-1)$ $= \frac{6 + a}{(x - 3)^2}$ $\frac{d^2y}{dx^2} = \frac{6 + a}{(x - 3)^3}(-2)$ $= -\frac{2(6 + a)}{(x - 3)^3}$ Where the graph is concave upwards, $\frac{d^2y}{dx^2} > 0$. Since $a < -6$, $6 + a < 0$. Therefore, $(x - 3)^3 > 0 \Rightarrow x > 3$

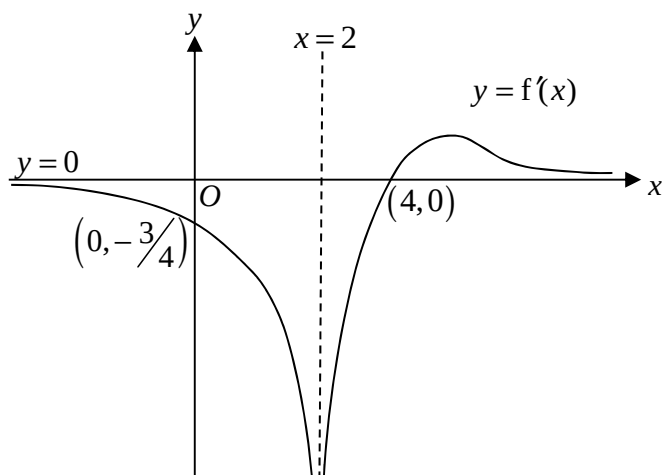
8 (b) (i)



8 (b) (ii)



8 (b) (iii)



9 (i)

Consider the volume of a can

$$\frac{2}{3}\pi r^3 + \pi r^2 h = k$$

$$\pi r h = \frac{k}{r} - \frac{2}{3}\pi r^2$$

$$h = \frac{k}{\pi r^2} - \frac{2}{3}r \quad \text{--- (1)}$$

Cost of a can,

$$C = 4(2\pi r^2) + 6(2\pi r h) + 6(\pi r^2)$$

$$= 14(\pi r^2) + 12(\pi r h)$$

$$= 14(\pi r^2) + \frac{12k}{r} - 8\pi r^2 \quad (\text{from (1)})$$

$$= 6\pi r^2 + \frac{12k}{r}$$

$$\frac{dC}{dr} = 12\pi r - \frac{12k}{r^2} = 0$$

$$r^3 = \frac{k}{\pi}$$

$$\therefore r = \sqrt[3]{\frac{k}{\pi}}$$

$$\frac{d^2C}{dr^2} = 12\pi + \frac{24k}{r^3} = 36\pi > 0$$

$$\text{or } \frac{d^2C}{dr^2} = 12\pi + \frac{24k}{r^3} > 0 \text{ since } r, k > 0$$

$$\text{When } r = \sqrt[3]{\frac{k}{\pi}},$$

$$h = \frac{k}{\pi \left(\sqrt[3]{\frac{k}{\pi}} \right)^2} - \frac{2}{3} \left(\sqrt[3]{\frac{k}{\pi}} \right)$$

$$= \sqrt[3]{\frac{k}{\pi}} - \frac{2}{3} \left(\sqrt[3]{\frac{k}{\pi}} \right) = \frac{1}{3} \left(\sqrt[3]{\frac{k}{\pi}} \right)$$

The cheapest can is manufactured when $r = \sqrt[3]{\frac{k}{\pi}}$ and $h = \frac{1}{3} \left(\sqrt[3]{\frac{k}{\pi}} \right)$.

9 (ii)

$$\text{Volume of hemisphere} = \frac{2}{3}\pi r^3 = \frac{128}{3}\pi$$

After 80 seconds, volume of water leaked

$$= 160 > \frac{128}{3}\pi$$

Hence we only consider the rate of change of the height of water in the **cylinder** after 80 seconds

Let H be the height in cm of the water in the can.

$$V = \pi r^2 H$$

$$= 16\pi H$$

$$\frac{dV}{dt} = 16\pi \frac{dH}{dt}$$

$$\frac{dH}{dt} = \frac{1}{16\pi} \frac{dV}{dt}$$

$$= \frac{-2}{16\pi}$$

$$= -\frac{1}{8\pi}$$

The height of the liquid is decreasing at $\frac{1}{8\pi}$ cm s⁻¹, 80 seconds after the start of the experiment.

10
(i)

$$\begin{aligned} \text{RHS} &= \frac{1}{\sqrt{1-p^2x^2}} - \sqrt{1-p^2x^2} \\ &= \frac{1-(1-p^2x^2)}{\sqrt{1-p^2x^2}} = \frac{p^2x^2}{\sqrt{1-p^2x^2}} = \text{LHS (shown)} \end{aligned}$$

Method 1

$$\begin{aligned} \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx &= \int_n^{\frac{1}{p}} \frac{1}{\sqrt{1-p^2x^2}} - \frac{p^2x^2}{\sqrt{1-p^2x^2}} \, dx \\ &= \int_n^{\frac{1}{p}} \frac{1}{\sqrt{1-p^2x^2}} \, dx + \frac{1}{2} \int_n^{\frac{1}{p}} x \cdot \frac{-2p^2x}{\sqrt{1-p^2x^2}} \, dx \\ &= \frac{1}{p} \int_n^{\frac{1}{p}} \frac{p}{\sqrt{1-(px)^2}} \, dx \\ &\quad + \left[\frac{1}{2} x \cdot \frac{\sqrt{1-p^2x^2}}{\frac{1}{2}} \right]_n^{\frac{1}{p}} - \frac{1}{2} \int_n^{\frac{1}{p}} \frac{\sqrt{1-p^2x^2}}{\frac{1}{2}} \, dx \\ &= \frac{1}{p} \left[\sin^{-1}(px) \right]_n^{\frac{1}{p}} + \left[0 - n\sqrt{1-p^2n^2} \right] - \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx \\ &= \frac{1}{p} \left[\sin^{-1}(1) - \sin^{-1}(pn) \right] - n\sqrt{1-p^2n^2} - \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx \\ &= \frac{1}{p} \left[\frac{\pi}{2} - \sin^{-1}(pn) \right] - n\sqrt{1-p^2n^2} - \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx \\ &= \frac{\pi}{2p} - \frac{1}{p} \sin^{-1}(pn) - n\sqrt{1-p^2n^2} - \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx \end{aligned}$$

Hence,

$$\begin{aligned} 2 \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx &= \frac{\pi}{2p} - \frac{1}{p} \sin^{-1}(pn) - n\sqrt{1-p^2n^2} \\ \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx &= \frac{\pi}{4p} - \frac{1}{2p} \sin^{-1}(pn) - \frac{1}{2} n\sqrt{1-p^2n^2} \end{aligned}$$

10 (i)

Method 2

$$\begin{aligned}
\int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx &= \int_n^{\frac{1}{p}} 1 \cdot \sqrt{1-p^2x^2} \, dx \\
&= \left[x\sqrt{1-p^2x^2} \right]_n^{\frac{1}{p}} - \int_n^{\frac{1}{p}} x \cdot \frac{1}{2\sqrt{1-p^2x^2}} \cdot (-2p^2x) \, dx \\
&= \left(0 - n\sqrt{1-p^2n^2} \right) + \int_n^{\frac{1}{p}} \frac{p^2x^2}{\sqrt{1-p^2x^2}} \, dx \\
&= -n\sqrt{1-p^2n^2} + \int_n^{\frac{1}{p}} \frac{1}{\sqrt{1-p^2x^2}} - \sqrt{1-p^2x^2} \, dx \\
&= -n\sqrt{1-p^2n^2} + \int_n^{\frac{1}{p}} \frac{1}{\sqrt{1-p^2x^2}} \, dx - \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx
\end{aligned}$$

Hence,

$$\begin{aligned}
2 \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx &= -n\sqrt{1-p^2n^2} + \int_n^{\frac{1}{p}} \frac{1}{\sqrt{1-p^2x^2}} \, dx \\
&\Rightarrow \int_n^{\frac{1}{p}} \sqrt{1-p^2x^2} \, dx \\
&= \frac{1}{2} \left[-n\sqrt{1-p^2n^2} + \frac{1}{p} \int_n^{\frac{1}{p}} \frac{p}{\sqrt{1-(px)^2}} \, dx \right] \\
&= -\frac{1}{2} n\sqrt{1-p^2n^2} + \frac{1}{2p} \left[\sin^{-1}(px) \right]_n^{\frac{1}{p}} \\
&= \frac{1}{2p} \left[\sin^{-1} 1 - \sin^{-1}(pn) \right] - \frac{1}{2} n\sqrt{1-p^2n^2} \\
&= \frac{1}{2p} \left[\frac{\pi}{2} - \sin^{-1}(pn) \right] - \frac{1}{2} n\sqrt{1-p^2n^2} \\
&= \frac{\pi}{4p} - \frac{1}{2p} \sin^{-1}(pn) - \frac{1}{2} n\sqrt{1-p^2n^2} \text{ (shown)}
\end{aligned}$$

10 (ii)

$$C_1 : \frac{x^2}{3} + y^2 = 1 \Rightarrow y^2 = 1 - \frac{1}{3}x^2$$

$$C_2 : x^2 + \frac{y^2}{3} = 1 \Rightarrow y^2 = 3 - 3x^2$$

At the point of intersection,

$$1 - \frac{1}{3}x^2 = 3 - 3x^2 \Rightarrow \left(3 - \frac{1}{3}\right)x^2 = 3 - 1$$

$$\Rightarrow \frac{8}{3}x^2 = 2$$

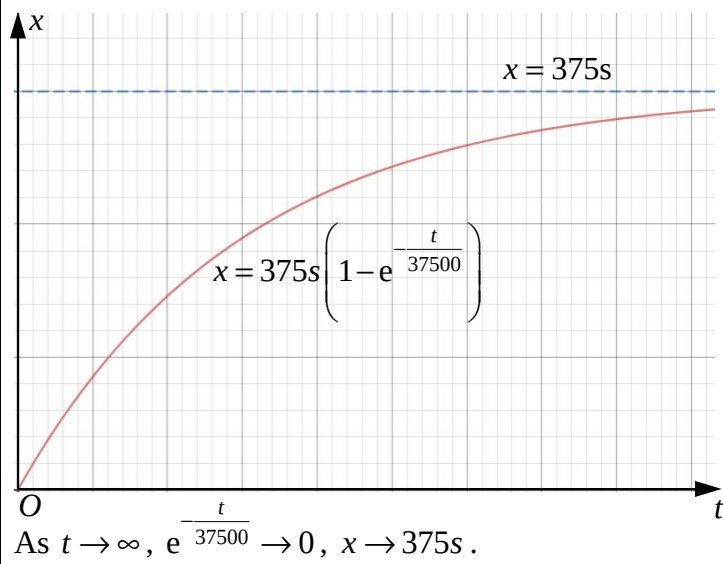
$$\Rightarrow x^2 = \frac{3}{4} \Rightarrow x = \frac{\sqrt{3}}{2} \quad (Q \ x > 0)$$

Therefore, area of shaded region is given by

$$\begin{aligned} & \int_{\frac{\sqrt{3}}{2}}^{\sqrt{3}} \sqrt{1 - \frac{1}{3}x^2} \, dx - \int_{\frac{\sqrt{3}}{2}}^1 \sqrt{3 - 3x^2} \, dx \\ &= \int_{\frac{\sqrt{3}}{2}}^{\sqrt{3}} \sqrt{1 - \left(\frac{1}{\sqrt{3}}\right)^2 x^2} \, dx - \sqrt{3} \int_{\frac{\sqrt{3}}{2}}^1 \sqrt{1 - x^2} \, dx \\ &= \frac{\pi}{4\left(\frac{1}{\sqrt{3}}\right)} - \frac{1}{2\left(\frac{1}{\sqrt{3}}\right)} \sin^{-1}\left(\frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2}\right) - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \sqrt{1 - \left(\frac{1}{\sqrt{3}}\right)^2 \left(\frac{\sqrt{3}}{2}\right)^2} \\ &\quad - \sqrt{3} \left[\frac{\pi}{4 \cdot 1} - \frac{1}{2 \cdot 1} \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \sqrt{1 - \left(\frac{\sqrt{3}}{2}\right)^2} \right] \\ &= \frac{\pi\sqrt{3}}{4} - \frac{\sqrt{3}}{2} \sin^{-1}\left(\frac{1}{2}\right) - \frac{\sqrt{3}}{4} \sqrt{\frac{3}{4}} \\ &\quad - \sqrt{3} \left[\frac{\pi}{4} - \frac{1}{2} \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \frac{\sqrt{3}}{4} \sqrt{\frac{1}{4}} \right] \\ &= -\frac{\sqrt{3}}{2} \cdot \frac{\pi}{6} - \frac{3}{8} + \frac{\sqrt{3}}{2} \cdot \frac{\pi}{3} + \frac{3}{8} = \frac{\pi\sqrt{3}}{12} \end{aligned}$$

11 (i)	Mass of free chlorine entering per minute $= \left(\frac{dx}{dt} \right)_{\text{in}} = 10s \text{ mg} = \frac{10s}{1000} = \frac{s}{100} \text{ g}$ Mass of free chlorine leaving per minute $= \left(\frac{dx}{dt} \right)_{\text{out}} = \frac{x}{375000} (10) \text{ g} = \frac{x}{37500} \text{ g}$ $\frac{dx}{dt} = \left(\frac{dx}{dt} \right)_{\text{in}} - \left(\frac{dx}{dt} \right)_{\text{out}}$ $= \frac{10s}{1000} - \frac{10x}{375000}$ $\Rightarrow \frac{dx}{dt} = \frac{375s - x}{37500} \text{ (shown)}$
	Assume that the mixture/concentration of free chlorine and water is uniform/homogeneous/well-mixed in the pool.
11 (ii)	$\frac{dx}{dt} = \frac{375s - x}{37500}$ $\int \frac{1}{375s - x} dx = \int \frac{1}{37500} dt$ $\Rightarrow -\ln 375s - x = \frac{t}{37500} + C$ $\Rightarrow -\ln(375s - x) = \frac{t}{37500} + C \text{ (since } 375s - x > 0)$ $\Rightarrow \ln(375s - x) = -\frac{t}{37500} - C$ $\Rightarrow 375s - x = e^{-\frac{t}{37500}} \cdot e^{-C}$ $\Rightarrow x = 375s - Ae^{-\frac{t}{37500}}, \text{ where } A = e^{-C}$ When $t = 0$, $x = 0$. $0 = 375s - A \Rightarrow A = 375s$ $\therefore x = 375s \left(1 - e^{-\frac{t}{37500}} \right)$
11 (iii)	In 60 minutes, $x = 375s \left(1 - e^{-\frac{60}{37500}} \right) = 0.599520s$ Amount of free chlorine in the pool after one hour is 0.600s or $375s \left(1 - e^{-\frac{1}{625}} \right)$ grams (or 600s milligrams).

11 (iv)



11 (last part)

$$375 < 375s < 1125$$

$$\Rightarrow 1 < s < 3$$